

Thermophysical Properties of the Lennard-Jones Fluid: Database and Data Assessment

Simon Stephan,^{*,†} Monika Thol,[‡] Jadran Vrabec,[§] and Hans Hasse[†]

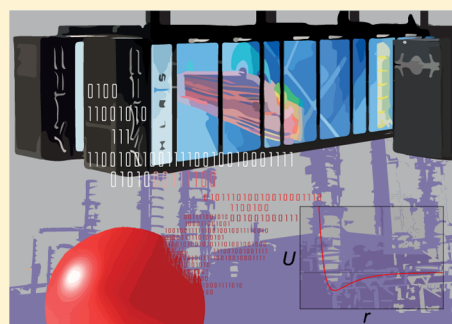
[†]Laboratory of Engineering Thermodynamics (LTD), TU Kaiserslautern, 67663 Kaiserslautern, Germany

[‡]Thermodynamics, Ruhr University Bochum, 44801 Bochum, Germany

[§]Thermodynamics and Process Engineering, TU Berlin, 10587 Berlin, Germany

Supporting Information

ABSTRACT: Literature data on the thermophysical properties of the Lennard-Jones fluid, which were sampled with molecular dynamics and Monte Carlo simulations, were reviewed and assessed. The literature data were complemented by simulation data from the present work that were taken in regions in which previously only sparse data were available. Data on homogeneous state points (for given temperature T and density ρ : pressure p , thermal expansion coefficient α , isothermal compressibility β , thermal pressure coefficient γ , internal energy u , isochoric heat capacity c_v , isobaric heat capacity c_p , Grüneisen parameter Γ , Joule–Thomson coefficient μ_{JT} , speed of sound w , Helmholtz energy a , and chemical potential) were considered, as well as data on the vapor–liquid equilibrium (for given T : vapor pressure p^s , saturated liquid and vapor densities ρ' and ρ'' , respectively, enthalpy of vaporization Δh_v , and as well as surface tension γ). The entire set of available data, which contains about 35 000 data points, was digitalized and included in a database, which is made available in the Supporting Information of this paper. Different consistency tests were applied to assess the accuracy and precision of the data. The data on homogeneous states were evaluated pointwise using data from their respective vicinity and equations of state. Approximately 10% of all homogeneous bulk data were discarded as outliers. The vapor–liquid equilibrium data were assessed by tests based on the compressibility factor, the Clausius–Clapeyron equation, and by an outlier test. Seven particularly reliable vapor–liquid equilibrium data sets were identified. The mutual agreement of these data sets is approximately $\pm 1\%$ for the vapor pressure, $\pm 0.2\%$ for the saturated liquid density, $\pm 1\%$ for the saturated vapor density, and $\pm 0.75\%$ for the enthalpy of vaporization—excluding the region close to the critical point.



INTRODUCTION

The assessment of experimental thermophysical property data is a well-established field in chemical engineering,^{1–3} especially for phase equilibrium data.^{4,5} For instance, consistency tests based on thermodynamic limits or balance equations are used to evaluate the quality of data sets or data points of a given thermophysical property. However, such approaches are not common in the analysis of thermophysical property data obtained by computer experiments like molecular dynamics (MD) or Monte Carlo (MC) simulations. Also, thermophysical data repositories^{6,7} have so far not addressed molecular simulation data of model fluids like the Lennard-Jones fluid. The present work therefore provides a coherent and consolidated database of thermophysical property data of the Lennard-Jones fluid.

The Lennard-Jones potential^{8,9} is of fundamental importance for the development of theories and methods in soft matter physics,^{10–12} and has been widely used since the early days of computer simulation.^{13–20} It is often taken as a benchmark for the validation of simulation codes and the test of new simulation techniques. Despite its simplicity, the Lennard-Jones model fluid yields a realistic representation of simple fluids.^{21,22} Due to its importance, it is sometimes even referred to as

Lennard-Jonesium^{23–25}—suggesting that it is viewed as a chemical element.²⁶ The Lennard-Jones potential is defined as the pairwise additive and spherically symmetric potential

$$u_{LJ}(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right] \quad (1)$$

where r is the distance between two particles. Its parameters ϵ and σ characterize the size of the particles and the magnitude of their dispersive attraction, respectively. Simulations are usually performed with a truncated potential in combination with a long-range correction.²⁷

To sample thermophysical properties of the Lennard-Jones fluid, computer experiments are carried out. In general, an obtained simulation result of a given observable x^{sim} will not agree with the true model value x^{mod} .²⁸ Like in experiments in the laboratory, errors occur also in computer experiments^{29–33} that can cause deviations between the exact true value x^{mod} and the value observed in the simulation x^{sim} . Both stochastic and systematic errors are usually present to some

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extent in computer experiments. Techniques to assess statistical errors are well established for computer simulations.^{27,30,34} It is more difficult to assess systematic simulation errors. As in laboratory experiments, round robin studies can be used for doing this, in which the same simulation task is carried out by different groups with different programs. It is known that the results from such studies generally differ by more than the combined statistical uncertainty of the individual data, e.g., the results obtained by repetition usually differ due to varying methods, simulation programs, etc.²⁹ Systematic errors may be a consequence of erroneous algorithms,³⁵ user errors, differences due to different simulation methods (e.g., MD and MC, phase equilibrium simulation methods, techniques for the determination of the chemical potential,³⁶ etc.). Systematic errors may furthermore be caused by finite size effects, erroneous evaluation of long-range interactions, insufficient equilibration or production periods, compilers, parallelization, and hardware architecture.²⁹ Last but not least, also typographical errors in publications have to be considered as a possible error source.

The detection and assessment of outliers in large data sets is a standard task in the field of data science^{37–39} and is widely applied to experimental data,^{40,41} but has to the best of our knowledge not yet been applied to thermophysical property data obtained by computer experiments with model fluids. In the present work, we use the terms accurate and precise as follows: accurate simulation results x^{sim} scatter around the true value x^{mod} without trend; precise means here that given simulation results x^{sim} are both accurate and exhibit a low scattering.

This article reviews and assesses molecular simulation data of the Lennard-Jones fluid. Approximately 35 000 data points were taken into account, including new simulation data from this work that were taken to complement the available data in regions that were only sparsely investigated in the literature. Vapor–liquid equilibrium (VLE) data and data on state points from the homogeneous regions were considered: For the VLE, the vapor pressure, the saturated densities, and the enthalpy of vaporization, and the surface tension were investigated; for homogeneous state points, the investigated properties are pressure, thermal expansion coefficient, isothermal compressibility, thermal pressure coefficient, internal energy, isochoric heat capacity, isobaric heat capacity, Grüneisen parameter, Joule–Thomson coefficient, speed of sound, Helmholtz energy, and chemical potential. The transport properties were not taken into account.

The data were assessed by consistency tests to provide an indication for their accuracy and precision. The entire database was digitalized, and is presented in a consistent form as a spreadsheet in the [Supporting Information](#). The database also contains flags that indicate whether a data point was identified as an outlier. All thermophysical property data are reported using reduced units with respect to the Lennard-Jones energy and size parameters ϵ and σ as well as the Boltzmann constant k_B .²⁷

■ MOLECULAR SIMULATION DATA OF THE LENNARD-JONES FLUID

Table 1 gives an overview on the thermophysical property data of the Lennard-Jones fluid from the literature. Only data on homogeneous states and vapor–liquid equilibria were considered in this work. Data on the second and third virial coefficients and the so-called characteristic curves^{42,43} were included in the database, but not further assessed regarding their accuracy. The vast majority of studies in the literature

report pVT and internal energy data for homogeneous state points (cf. Table 1). The chemical potential and higher-order derivatives like speed of sound, heat capacities, or isothermal compressibility were less frequently investigated. Wherever statistical uncertainties were reported in the literature, they were also included in the database.

The simulation method proposed by Lustig^{84,175} provides Helmholtz energy derivatives with respect to the density and the inverse temperature simultaneously from a single simulation run. This technique can be combined with Widom's test particle insertion method¹⁷⁶ for the determination of the Helmholtz energy itself. This approach was applied to the Lennard-Jones fluid by Thol et al.⁹⁰ and Köster et al.^{91,92} in different fluid regions. The simulations of Thol et al.⁹⁰ were restricted to stable fluid states below $T = 9$, while the simulation data of Köster et al.^{91,92} focused on the high-density region close to the freezing line. These two data sets are complemented in the present work by data on 655 additional homogeneous state points (cf. Figure 1) that were also sampled with the Lustig formalism (LF). The new data were taken to get more information on regions in which literature data were comparatively sparse. For the metastable region, only pressure, chemical potential, and internal energy are reported here. The entirety of the simulation results of refs 90–92 and from this work is referred to as LF data set in the following. All of them were sampled with the simulation program *ms2*¹⁷⁷ using similar simulation settings. Details on the simulation procedure are given in [Appendix](#).

Results on the VLE properties of the Lennard-Jones fluid have been reported many times in the literature (cf. Table 1). A total of 45 VLE data sets were found in the progress of this work. The simulation methods used in these studies can be separated into two types: simulations in which the liquid and vapor phases are simulated in two separate volumes that are suitably coupled and simulations in which the liquid and vapor phases coexist in a single volume that also contains the interface. The first are referred to as indirect methods, and the latter as direct methods in the following. Both VLE simulation types provide in general vapor pressure p^s , saturated densities of the liquid and vapor phases ρ' and ρ'' , respectively, and enthalpy of vaporization Δh_v at a given temperature T . However, often only a subset of these properties was reported in the literature (cf. Table 1). Simulation techniques that belong to the indirect type¹⁷⁸ are the Gibbs ensemble method,^{100,101} the Grand equilibrium method,¹⁷⁹ the NpT + test particle method,¹⁸⁰ the Gibbs–Duhem integration,¹⁸¹ and the Wang–Landau method.¹⁸² Additional information on interfacial properties can be obtained with the direct method.^{183,184} Data on the surface tension of the Lennard-Jones fluid are also included in the database (cf. Table 1). The Grand equilibrium method was employed in the present work to obtain an extended VLE data set for the Lennard-Jones fluid. Details are given in [Appendix](#).

Data on the triple point^{18,46,48,60,61,70,71,73,75,78,100,104,107,113,116,119,120,122,167,186,189–205} and the critical point^{18,46,48,60,61,70,71,73,75,78,100,104,107,113,116,119,120,122,167,186,189–205} of the Lennard-Jones fluid have been reported

many times. Numeric values of the critical point are summarized in Table 2. The critical pressure has been less frequently reported than the critical temperature and density. The critical parameters of Lennard–Jones fluid reported in the literature scatter over a large range—especially regarding the temperature. However, 13 critical temperature and density data points (refs 71 104 119 120 122 191 197 198 200, and 202) are in fairly good agreement, even though most of them do not agree within

Table 1. Computer Experiment Data of the Lennard-Jones Fluid from the Literature^a

authors	year	#	T	authors	year	#	T
<i>pVT</i> data				<i>pVT</i> data			
Wood and Parker ¹⁵	1957	13	2.74	Köster et al. ⁹¹	2017	45	1.01–30
Fickett and Wood ⁴⁴	1960	23	1.92–177.95	Köster et al. ^{91,92}	2017	65	1.01–30
McDonald and Singer ⁴⁵	1967	28	0.72–1.24	Ustinov ^{93,94}	2017	232	0.76–1.14
McDonald and Singer ¹⁹	1967	48	1.45–3.53	Schultz and Kofke ⁹⁵	2018	404	0.68–2272
Verlet and Levesque ⁴⁶	1967	7	1.05–2.74	this work	2019	655	0.7–90
Verlet ¹⁸	1967	39	0.59–4.63	Internal Energy <i>u</i>			
Wood ⁴⁷	1968	41	1.06–100	Wood and Parker ¹⁵	1957	13	2.74
Hansen and Verlet ²⁰	1969	25	0.75–1.15	McDonald and Singer ⁴⁵	1967	27	0.72–1.24
Levesque and Verlet ⁴⁸	1969	25	0.72–3.67	McDonald and Singer ¹⁹	1967	48	1.45–3.53
McDonald and Singer ⁴⁹	1969	28	0.72–1.24	Verlet ¹⁸	1967	39	0.59–4.63
Hansen ⁵⁰	1970	9	2.74–100	Verlet and Levesque ⁴⁶	1967	7	1.05–2.74
McDonald and Woodcock ⁵¹	1970	6	0.75–2.33	Wood ⁴⁷	1968	41	1.06–100
Toxvaerd and Praetgaard ⁵²	1970	8	1.35	Levesque and Verlet ⁴⁸	1969	25	0.72–3.67
Weeks et al. ⁵³	1971	5	0.75–1.35	McDonald and Singer ⁴⁹	1969	28	0.72–1.24
McDonald and Singer ⁵⁴	1972	56	0.55–1.24	Hansen ⁵⁰	1970	9	2.74–100
Schofield ⁵⁵	1973	6	0.73–1.1	McDonald and Woodcock ⁵¹	1970	6	0.75–2.33
Street et al. ⁵⁶	1974	80	0.75–3.05	Weeks et al. ⁵³	1971	5	0.75–1.35
Adams ⁵⁷	1975	12	2–4	McDonald and Singer ⁵⁴	1972	56	0.55–1.24
Adams ⁵⁸	1976	16	1–1.2	Street et al. ⁵⁶	1974	80	0.75–3.05
Carley ⁵⁹	1977	11	1.35	Adams ⁵⁷	1975	12	2–4
Adams ⁶⁰	1979	31	1.15–1.35	Adams ⁵⁸	1976	16	1–1.2
Nicolas et al. ⁶¹	1979	108	0.48–6.01	Torrie and Valleau ⁹⁶	1977	7	0.092–1.35
Ree ⁶²	1980	11	0.81–2.7	Adams ⁶⁰	1979	31	1.15–1.35
Yao et al. ⁶³	1982	12	1.15–1.25	Nicolas et al. ⁶¹	1979	108	0.48–6.01
Powles et al. ⁶⁴	1982	37	0.7–1.41	Ree ⁶²	1980	11	0.81–2.7
Lucas ⁶⁵	1986	10	0.79–1.83	Yao et al. ⁶³	1982	12	1.15–1.25
Shaw ⁶⁶	1988	265	0.6–136.25	Lucas ⁶⁵	1986	10	0.79–1.83
Adachi et al. ⁶⁷	1988	328	0.7–2.95	Shaw ⁶⁶	1988	265	0.59–136.25
Baranyai et al. ⁶⁸	1989	12	0.75–1.5	Baranyai et al. ⁶⁸	1989	18	0.75–1.5
Saager and Fischer ⁶⁹	1990	38	0.57–4	Saager and Fischer ⁶⁹	1990	38	0.57–4
Sowers and Sandler ⁷⁰	1991	60	1.35–6	Sowers and Sandler ⁷⁰	1991	54	1.35–6
Lotfi et al. ⁷¹	1992	19	0.7–1.3	Lotfi et al. ⁷¹	1992	19	0.7–1.3
Giaquinta et al. ⁷²	1992	12	0.75–1.15	Giaquinta et al. ⁷²	1992	12	0.75–1.15
Johnson et al. ⁷³	1993	199	0.7–6	Johnson et al. ⁷³	1993	199	0.7–6
Kolafa et al. ⁷⁴	1993	43	0.72–4.85	Kolafa et al. ⁷⁴	1993	43	0.72–4.85
Miyano ⁷⁵	1993	112	0.45–100	Miyano ⁷⁵	1993	112	0.45–100
Kolafa and Nezbeda ⁷⁶	1994	13	0.81–10	Kolafa and Nezbeda ⁷⁶	1994	13	0.81–10
Lustig ⁷⁷	1994	2	1.18	Lustig ⁷⁷	1994	2	1.18
Mecke et al. ⁷⁸	1996	12	1.32–1.34	Mecke et al. ⁷⁸	1996	12	1.32–1.34
Roccatano et al. ⁷⁹	1998	10	1.4–10	Roccatano et al. ⁷⁹	1998	10	1.4–10
Meier ⁸⁰	2002	351	0.7–6	Meier ⁸⁰	2002	351	0.7–6
Linhardt et al. ⁸¹	2005	108	0.7–1.2	Baidakov et al. ⁸³	2008	201	0.35–2
Morsali et al. ⁸²	2007	13	5.01	May and Mausbach ⁸⁵	2012	218	0.68–6.17
Baidakov et al. ⁸³	2008	208	0.35–2	Yigzawe and Sadus ^{86,87}	2012	346	1.31–2.62
Lustig ⁸⁴	2011	8	1.02–3	Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05
May and Mausbach ⁸⁵	2012	205	0.69–6.17	Thol et al. ⁹⁰	2016	197	0.7–9
Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62	Deiters and Neumaier ^{43,97}	2016	255	0.67–22.54
Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05	Köster et al. ⁹¹	2017	45	1.01–30
Thol et al. ⁹⁰	2016	197	0.7–9	Köster et al. ^{91,92}	2017	65	1.01–30
Deiters and Neumaier ⁴³	2016	255	0.67–22.54	Ustinov ^{93,94}	2017	232	0.76–1.14

Table 1. continued

authors	year	#	T	authors	year	#	T
Internal Energy u				VLE $p^s, \rho', \rho'', \Delta h_v$			
Schultz and Kofke ⁹⁵	2018	404	0.68–2272	this work ($p^s, \rho', \rho'', \Delta h_v$)	2019	124	0.69–1.28
this work	2019	655	0.7–90	Vapor–Liquid Surface Tension γ			
VLE $p^s, \rho', \rho'', \Delta h_v$				Lee et al. ⁹⁸	1974	5	0.7–1.2
Hansen and Verlet ²⁰ ($p^s, \rho', \rho'', \Delta h_v$)	1969	8	0.75–1.15	Miyazaki et al. ¹³⁹	1976	1	0.7
Lee et al. ⁹⁸ (ρ', ρ'')	1974	10	0.7–1.2	Chapela et al. ⁹⁹	1977	5	0.7–1.27
Adams ⁵⁸ ($p^s, \rho', \rho'', \Delta h_v$)	1976	44	0.6–1.1	Nijmeijer et al. ¹⁰²	1988	1	0.92
Chapela et al. ⁹⁹ (ρ', ρ'')	1977	6	0.7–0.84	Holcomb et al. ¹⁰⁵	1993	3	0.72–1.13
Adams ⁶⁰ ($p^s, \rho', \rho'', \Delta h_v$)	1979	11	1.15–1.30	Mecke et al. ¹⁴⁰	1997	3	0.7–1.1
Panagiotopoulos ¹⁰⁰ ($p^s, \rho', \rho'', \Delta h_v$)	1987	30	0.75–1.3	Guo et al. ¹¹¹	1997	5	0.75–1.25
Panagiotopoulos et al. ¹⁰¹ ($p^s, \rho', \rho'', \Delta h_v$)	1988	30	0.75–1.3	Guo and Lu ¹¹²	1997	4	0.75–1.15
Nijmeijer et al. ¹⁰² (ρ', ρ'')	1988	2	0.92	Trokhymchuk and Alejandre ¹¹⁴	1999	6	0.72–1.27
Smit and Frenkel ¹⁰³ ($p^s, \rho', \rho'', \Delta h_v$)	1989	20	1.15–1.3	Anisimov et al. ¹¹⁵	1999	6	0.75–1
Lotfi et al. ⁷¹ ($p^s, \rho', \rho'', \Delta h_v$)	1992	52	0.7–1.3	Potoff and Panagiotopoulos ¹¹⁶	2000	18	0.95–1.31
Kofke ¹⁰⁴ ($p^s, \rho', \rho'', \Delta h_v$)	1993	80	0.74–1.32	Baidakov et al. ¹¹⁷	2000	7	0.72–1.23
Holcomb et al. ¹⁰⁵ (ρ', ρ'')	1993	6	0.72–1.13	Chen et al. ²⁴	2001	3	0.7–0.8
Agrawal and Kofke ¹⁰⁶ ($p^s, \rho', \rho'', \Delta h_v$)	1995	52	0.68–0.74	Baidakov et al. ¹²³	2002	6	0.72–1.22
Hunter and Reinhardt ¹⁰⁷ (ρ', ρ'')	1995	39	1–1.35	Errington ¹²⁵	2003	4	0.7–1.3
Sadus and Prausnitz ¹⁰⁸ ($p^s, \rho', \rho'', \Delta h_v$)	1996	30	1–1.25	Baidakov et al. ¹²⁹	2007	12	0.5–1.2
Mecke et al. ¹⁰⁹ (ρ', ρ'')	1997	6	0.7–1.1	Shen et al. ¹⁴¹	2007	10	0.7–1.1
Pláčkov and Sadus ¹¹⁰ ($p^s, \rho', \rho'', \Delta h_v$)	1997	55	0.95–1.27	Janeček et al. ¹³¹	2009	4	0.72–1.25
Guo et al. ¹¹¹ (ρ', ρ'')	1997	10	0.75–1.25	Galliero et al. ¹³²	2009	13	0.7–1.3
Guo and Lu ¹¹² (ρ', ρ'')	1997	8	0.75–1.15	Galliero et al. ¹³²	2009	6	0.7–1.2
Martin and Siepmann ¹¹³ (p^s, ρ', ρ'')	1998	18	0.75–1.18	Werth et al. ¹⁴²	2013	7	0.7–1.25
Trokhymchuk and Alejandre ¹¹⁴ (p^s, ρ', ρ'')	1999	18	0.72–1.27	Martínez-Ruiz et al. ¹³⁴	2014	7	0.7–1.1
Anisimov et al. ¹¹⁵ (p^s, ρ', ρ'')	1999	18	0.75–1	Janeček et al. ¹³⁵	2017	8	0.7–1.25
Potoff and Panagiotopoulos ¹¹⁶ (ρ', ρ'')	2000	36	0.95–1.31	Werth et al. ¹³⁷	2017	9	0.72–1.24
Baidakov et al. ¹¹⁷ (p^s, ρ', ρ'')	2000	21	0.72–1.23	Stephan and Hasse ¹³⁸	2019	13	0.69–1.29
Okumura and Yonezawa ^{118,119} (p^s, ρ', ρ'')	2000	39	0.7–1.3	SLE			
Shi and Johnson ¹²⁰ (ρ', ρ'')	2001	26	1.15–1.27	Hansen and Verlet ²⁰	1969	4	0.75–2.74
Chen et al. ²⁴ (ρ', ρ'')	2001	6	0.7–0.8	Hansen ⁵⁰	1970	6	2.74–100
Okumura and Yonezawa ^{121,122} (p^s, ρ', ρ'')	2001	30	1.25–1.32	Agrawal and Kofke ¹⁰⁶	1995	37	0.69–274
Baidakov et al. ¹²³ ($\rho', \rho'', \text{spinodal}$)	2002	14	0.72–1.23	van der Hoef ¹⁴³	2000	corr. ^b	0.1–2.0
Kioupis et al. ¹²⁴ ($p^s, \rho', \rho'', \Delta h_v$)	2002	40	1.03–1.3	Barroso and Ferreira ¹⁴⁴	2002	18	0.69–4.5
Errington ¹²⁵ (p^s, ρ', ρ'')	2003	12	0.7–1.3	Morris and Song ¹⁴⁵	2002	12	0.72–2.65
Errington ^{126,127} (p^s, ρ', ρ'')	2003	39	0.7–1.3	Errington ¹⁴⁶	2004	2	0.75–2
Stoll et al. ¹²⁸ ($p^s, \rho', \rho'', \Delta h_v$)	2003	44	0.73–1.26	McNeil-Watson and Wilding ¹⁴⁷	2006	34	0.72–83
Baidakov et al. ¹²⁹ (p^s, ρ', ρ'')	2007	36	0.5–1.2	Mastny and Pablo ¹⁴⁸	2007	5	1–20
Betancourt-Cárdenas et al. ¹³⁰ ($p^s, \rho', \rho'', \Delta h_v$)	2008	35	0.7–1.27	Ahmed and Sadus ^{149,150}	2009	5	0.8–2.74
Janeček ¹³¹ ($p^s, \rho', \rho'', \Delta h_v$)	2009	16	0.72–1.25	Sousa et al. ¹⁵¹	2012	10	0.75–5
Galliero et al. ¹³² (ρ', ρ'')	2009	38	0.7–1.3	Köster et al. ⁹¹	2017	8	1.3–30
Sadus ¹³³ (p^s)	2012	14	0.7–1.29	Schultz and Kofke ⁹⁵	2018	corr. ^b	0.68–2272
Mick et al. ²³ ($p^s, \rho', \rho'', \Delta h_v$)	2013	36	0.75–1.3	Isochoric Heat Capacity c_v			
Martínez-Ruiz et al. ¹³⁴ (p^s, ρ', ρ'')	2014	21	0.7–1.1	Wood and Parker ¹⁵	1957	11	2.74
Janeček et al. ¹³⁵ ($p^s, \rho', \rho'', \Delta h_v$)	2017	31	0.7–1.25	McDonald and Singer ⁴⁵	1967	26	0.72–1.24
Janeček et al. ^{135,136} ($p^s, \rho', \rho'', \Delta h_v$)	2017	24	0.8–1.25	McDonald and Singer ¹⁹	1967	48	1.45–3.53
Werth et al. ¹³⁷ ($p^s, \rho', \rho'', \Delta h_v$)	2017	36	0.72–1.24	Adams ⁵⁷	1975	12	2–4
Stephan and Hasse ¹³⁸ (p^s, ρ', ρ'')	2019	39	0.69–1.29	Adams ⁶⁰	1979	31	1.15–1.35
				Saager et al. ^{152,153}	1990	12	1.1–1.35
				Boda et al. ¹⁵⁴	1996	9	1.31–2
				Roccatano et al. ⁷⁹	1998	10	1.4–10

Table 1. continued

authors	year	#	T	authors	year	#	T
Isochoric Heat Capacity c_v				Thermal Pressure Coefficient γ			
Meier ^{80,155}	2002	327	0.7–6	Lustig ⁷⁷	1994	2	1.18
Baidakov et al. ⁸³	2008	208	0.35–2	Meier ⁸⁰	2002	326	0.7–6
May and Mausbach ^{85,156}	2012	218	0.68–6.17	Morsali et al. ⁸²	2007	13	3.76
Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62	May and Mausbach ^{85,156}	2012	205	0.69–6.17
Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05	Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62
Thol et al. ⁹⁰	2016	197	0.7–9	Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05
Köster et al. ⁹¹	2017	45	1.01–30	Thol et al. ⁹⁰	2016	197	0.7–9
Köster et al. ^{91,92}	2017	65	1.01–30	Köster et al. ⁹¹	2017	45	1.01–30
this work	2019	515	0.7–90	Köster et al. ^{91,92}	2017	65	1.01–30
Isobaric Heat Capacity c_p				this work	2019	515	0.7–90
Boda et al. ¹⁵⁴	1996	41	0.65–1.9	Speed of Sound w			
Lustig ⁸⁴	2011	6	1.02–3	Meier ⁸⁰	2002	349	0.7–6
May and Mausbach ^{85,156}	2012	202	0.69–6.17	Lustig ⁸⁴	2011	8	1.02–3
Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62	May and Mausbach ^{85,156}	2012	205	0.69–6.17
Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05	Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62
Thol et al. ⁹⁰	2016	197	0.7–9	Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05
Köster et al. ⁹¹	2017	45	1.01–30	Thol et al. ⁹⁰	2016	197	0.7–9
Köster et al. ^{91,92}	2017	65	1.01–30	Köster et al. ⁹¹	2017	45	1.01–30
this work	2019	515	0.7–90	Köster et al. ^{91,92}	2017	65	1.01–30
Grüneisen Parameter Γ				this work	2019	515	0.7–90
Emampour et al. ¹⁵⁷	2011	26	1.2–1.8	Joule–Thomson Coefficient μ_T			
Mausbach and May ¹⁵⁸	2014	212	0.69–6.17	Lustig ⁸⁴	2011	8	1.02–3
Thol et al. ⁹⁰	2016	197	0.7–9	May and Mausbach ^{85,156}	2012	205	0.69–6.17
Mausbach et al. ¹⁵⁹	2016	110	0.72–9	Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62
Köster et al. ⁹¹	2017	45	1.01–30	Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05
Köster et al. ^{91,92}	2017	65	1.01–30	Thol et al. ⁹⁰	2016	197	0.7–9
this work	2019	515	0.7–90	Köster et al. ⁹¹	2017	45	1.01–30
Thermal Expansion Coefficient α				Köster et al. ^{91,92}	2017	65	1.01–30
McDonald and Singer ⁴⁵	1967	20	0.72–1.24	this work	2019	515	0.7–90
Adams ⁵⁷	1975	12	2–4	Entropic Properties, a , μ			
Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62	Levesque and Verlet ⁴⁸ (a)	1969	8	1.35
Thol et al. ⁹⁰	2016	197	0.7–9	Hansen and Verlet ²⁰ (a)	1969	25	0.75–1.15
Köster et al. ⁹¹	2017	45	1.01–30	Weeks et al. ⁵³ (a)	1971	5	0.75–1.35
Köster et al. ^{91,92}	2017	65	1.01–30	Weeks et al. ¹⁶⁰ (a)	1971	27	0.75–1.35
this work	2019	515	0.7–90	Adams ⁵⁷ (μ)	1975	12	2–4
Isothermal Compressibility β				Torrie and Valleau ⁹⁶ (a)	1977	16	0.75–2.74
McDonald and Singer ⁴⁵	1967	22	0.72–1.24	Adams ⁶⁰ (μ)	1979	31	1.15–1.35
Adams ⁵⁷	1975	12	2–4	Yao et al. ⁶³ (μ)	1982	12	1.15–1.25
Adams ⁶⁰	1979	31	1.15–1.35	Powles et al. ⁶⁴ (μ)	1982	37	0.7–1.41
Lotfi et al. ⁷¹	1992	19	0.7–1.3	Panagiotopoulos et al. ¹⁰¹ (μ)	1988	18	0.75–1.3
Lustig ⁷⁷	1994	2	1.18	Baranyai and Evans ⁶⁸ (a)	1989	6	1.15
Morsali et al. ⁸²	2007	13	3.76	Lotfi et al. ⁷¹ (μ)	1992	19	0.75–1.3
May and Mausbach ^{85,156}	2012	205	0.69–6.17	Han ¹⁶¹ (μ)	1992	3	1.2
Yigzawe and Sadus ^{86,87}	2012	406	1.3–2.62	Kolafa et al. ⁷⁴ (μ)	1993	7	1.2–1.45
Mairhofer and Sadus ^{88,89}	2013	282	1.36–3.05	Lustig ⁷⁷ (μ)	1994	2	1.18
Thol et al. ⁹⁰	2016	197	0.7–9	Cuadros et al. ¹⁶² (a)	1996	269	0.7–2.6
Köster et al. ⁹¹	2017	45	1.01–30	Hong and Jhon ¹⁶³ (a)	1997	36	0.59–2.89
Köster et al. ^{91,92}	2017	65	1.01–30	Hong and Jang ¹⁶⁴ (a)	2003	22	0.59–2.85
this work	2019	515	0.7–90	Thol et al. ⁹⁰ (a , μ)	2016	197	0.7–9
Thermal Pressure Coefficient γ				Köster et al. ^{91,92} (a , μ)	2017	65	1.01–30
McDonald and Singer ⁴⁵	1967	20	0.72–1.24	Ustinov ^{93,94} (μ)	2017	232	0.76–1.14
Adams ⁵⁷	1975	12	2–4	this work (a , μ)	2019	655	0.7–90

Table 1. continued

authors	year	#	T	authors	year	#	T
Virial Coefficients B, C				Ideal Curves ID, BL, JTI, JI ^c			
Bird et al. ¹⁶⁵ (C)	1950	74	0.7–400	Heyes and Llaguno ¹⁷² (JTI)	1992	16	1.15–7.1
Hirschfelder et al. ¹⁶⁶ (B, C)	1954	156	0.3–400	Colina and Müller ¹⁷³ (JTI)	1999	24	1.12–6.4
Barker et al. ¹⁶⁷ (B, C)	1966	66	0.625–20	Kioupis et al. ¹²⁴ (JTI)	2002	29	1.3–6
Nicolas et al. ⁶¹ (B)	1979	33	0.625–20	Vrabec et al. ¹⁷⁴ (JTI)	2005	18	1.2–6.3
Sun and Teja ¹⁶⁸ (B, C)	1996	302	0.39–6.1	Yigzawe and Sadus ⁸⁷ (JTI)	2013	9	1.31–2.62
Shaul et al. ¹⁶⁹ (B, C)	2010	22	0.7–2	Deiters and Neumaier ^{43,97} (ID, BL, JTI, JI)	2016	43	0.01–6.4
Wheatley ^{170,171} (B, C)	2013	100	0.1–1000				

^aThe data are sorted chronologically for each property. # is the number of data points, and T the investigated temperature range. ^bThe authors provided no numerical data but a correlation that was parametrized to computer experiment data. ^cID: classical ideal curve, BL: Boyle curve, JTI: Joule–Thomson inversion curve, JI: Joule inversion curve.

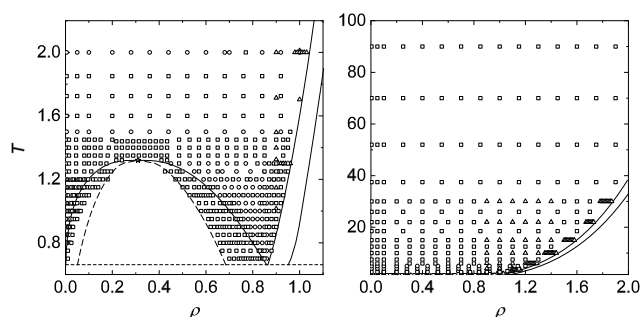


Figure 1. Overview of 963 state points that were studied with the Lustig formalism^{84,175} by different authors. Circles: Thol et al.;⁹⁰ triangles: Köster et al.;^{91,92} squares: this work. Data for the Helmholtz energy and its density and inverse temperature derivatives up to second order are available for the stable state points. For the metastable state points, the derivatives are available up to first order.

their combined statistical uncertainties. The critical temperature reported by refs 71 104 119 120 122 191 197 198 200, and 202 scatters in a range of $T_c = 1.31$ – 1.326 and critical density in a range of $\rho_c = 0.304$ – 0.318 . The reported critical density obtained from computer experiment reported by refs 71 104 119 120 122 191 197 198 200, and 202 cluster around two different values: $\rho_c = 0.305$ and 0.316 . Other data points were discarded since they significantly deviate from this entity. The critical pressure reported by refs 60 119 122 192, and 197 scatters in the range $p_c = 0.125$ – 0.135 .

In the following, the critical point is presumed to be located at $T_c = 1.321 \pm 0.007$, $\rho_c = 0.316 \pm 0.005$, and $p_c = 0.129 \pm 0.005$. The stated uncertainties were estimated from the standard deviation of the critical parameters reported by refs 71 104 119 120 122 191 197 198 200, and 202 for the critical temperature and density and refs 60 119 122 192, and 197 for the critical pressure.

The solid–fluid equilibrium of the Lennard-Jones fluid is used in this study only to delimit the fluid region, as molecular simulation results beyond the freezing line were excluded regarding the assessment. Solid–fluid equilibria have been investigated multiple times (cf. Table 1). The correlation for the freezing line by Köster et al.⁹¹ was employed here.

DATABASE OF THERMOPHYSICAL PROPERTIES

All thermophysical property data (approximately 35 000 data points) considered in this work are summarized in a consistent form in an .xls spreadsheet in the Supporting Information of this work. All data are sorted by thermophysical properties in

that spreadsheet. All numerical values are given in a consistent form, i.e., as residual properties with respect to the ideal gas and in the standard Lennard-Jones units. Information on the statistical uncertainties were adopted from publications—wherever such information was reported, which is unfortunately not always the case. The methods for the estimation of statistical uncertainties differ significantly among the publications. Sometimes, statistical uncertainties were reported, but no description on how such were obtained is given. Due to this heterogeneity, statistical uncertainties could unfortunately not be used for the assessment of data points in this work.

The database furthermore contains notes on pitfalls regarding the conversion of the primary literature data to the format used in the database. Furthermore, known misprints in publications from the literature were corrected, e.g., refs 90 92, and 155. A survey of these misprints is also given. Each data point possesses an additional mark, which indicates whether the data point was found to be an outlier or not, according to the assessment described in the following.

ASSESSMENT OF MOLECULAR SIMULATION DATA

Assessment of Data on Homogeneous States. Thermophysical property data of the Lennard-Jones fluid reported in the literature (cf. Table 1) for homogeneous state points were assessed by an equation of state (EOS) consistency test. This consistency test is based on the statistical method proposed by Rousseeuw and Croux²⁰⁶ for the detection of outliers that was adopted and extended for the purposes of the present work. The EOS test evaluates each data point individually by comparing it with the most accurate equations of state of the Lennard-Jones fluid available in the literature and computer experiment data in its vicinity. The EOS test is designed in a way to make a binary decision: either a data point is an outlier or not.

The relative deviation δx of a given data point $x_{DP}(T, \rho)$ from a given EOS at the same temperature and density was computed as

$$\delta x = \frac{x_{DP} - x_{EOS}}{x_{EOS}} \quad (2)$$

where x is one of the following homogeneous bulk phase properties: pressure p , internal energy u , speed of the sound w , thermal pressure coefficient γ , Grüneisen parameter Γ , thermal expansion coefficient α , isothermal compressibility β , Joule–Thomson coefficient μ_{JT} , isochoric heat capacity c_v , isobaric

Table 2. Critical Point Data for the Lennard-Jones Fluid Obtained by Computer Experiment Reported in the Literature^a

publication	year	T_c	ρ_c	P_c
Barker et al. ¹⁶⁷	1966	1.291	0.547	0.249
Verlet and Levesque ⁴⁶	1967	1.26	0.316	0.118
Verlet ¹⁸	1967	1.32	0.31	0.12276
Kim et al. ¹⁹⁴	1969	1.35	0.3	0.14
		1.28	0.29	0.13
Levesque and Verlet ⁴⁸	1969	1.36(3)	0.36(3)	0.151776
Sung and Chandler ²⁰⁰	1974	1.348	0.349	0.148
Adams ⁶⁰	1979	1.30(2)	0.33(3)	0.13(2)
Panagiotopoulos ¹⁰⁰	1987	<1.35	0.31(2)	
Meroni et al. ²¹⁷	1990	1.3317	0.332	
Smit ¹⁹⁸	1992	1.316(6)	0.304(6)	
Lotfi et al. ⁷¹	1992	1.31	0.314	
Kofke ¹⁰⁴	1993	1.321(4)	0.306(1)	
		1.324(12)	0.305(2)	
Valleau ^{78,218}	1994	1.328(3)		
Hunter and Reinhardt ¹⁰⁷	1995	1.32		
Mecke et al. ⁷⁸	1996	1.328(3)	0.3107	
Mecke et al. ¹⁰⁹	1997	1.32521		
Caillol ¹⁹¹	1998	1.326(2)	0.316(2)	0.1472(3)
Martin and Siepmann ¹¹³	1998	1.294(9)	0.311(6)	0.1144
		1.311(9)		
Potoff and Panagiotopoulos ¹⁹⁷	1998	1.3120(7)	0.316(1)	0.1279(6)
Potoff and Panagiotopoulos ¹¹⁶	2000	1.311(2)		
		1.3120(7)	0.316(2)	
Okumura and Yonezawa ¹¹⁹	2000	1.313(2)	0.304(2)	0.125(1)
Shi and Johnson ¹²⁰	2001	1.3241(9)	0.3165(7)	
		1.3145(2)	0.316(1)	
Dunikov et al. ¹⁹²	2001	1.350(5)	0.310(5)	0.126(8)
Okumura and Yonezawa ¹²²	2001	1.3207(4)	0.316(1)	0.1288(5)
Pérez-Pellitero et al. ²¹⁹	2006	1.3123(6)	0.3174(6)	
		1.3126(6)	0.3174(6)	
		1.313(1)	0.317(1)	
Betancourt-Cárdenas et al. ¹³⁰	2008	1.302(6)	0.311(2)	0.12(3)
Kulinskii ²⁰¹	2013	1.333	0.325	0.122
Heyes and Woodcock ²⁰³	2013	1.3365(5)		0.1405(2)
Heyes ²⁰²	2015	1.316(1)	0.309(6)	
Lishchuk and Fischer ²⁰⁵	2018	1.31766		

^aThe numbers in parentheses indicate the reported uncertainties of the last decimal digits.

heat capacity c_p , Helmholtz energy divided by the temperature $\bar{a} = a/T$, or chemical potential μ .

Comprehensive comparisons of EOS for the Lennard-Jones fluid have been carried out recently.^{90,207} Six of the most accurate EOS were used here: Johnson et al.,⁷³ Kolafa and Nezbeda,⁷⁶ Lafitte et al.,¹¹ Mecke et al.,^{78,208} Stephan et al.,²⁰⁷ and Thol et al.⁹⁰ (alphabetically). However, also these EOS have deficiencies, e.g., the EOS of Thol et al.⁹⁰ exhibits an unrealistic behavior in the two-phase region, that of Mecke et al.^{78,208} shows large deviations in the high-density region close to the freezing line, that of Stephan et al.²⁰⁷ is less accurate in the high-temperature region, the EOS of Kolafa and Nezbeda⁷⁶ produces distorted characteristic curves,⁴³ and the one of Lafitte et al.¹¹ is less accurate in the low-density gas region. The EOS of Johnson et al.⁷³ is overall less precise than the ones mentioned before. To take these deficiencies into account in evaluating an individual data point, only the four EOS were considered that yielded the smallest relative deviation δx_j for a given data point. The four best EOS were identified and selected for each data point individually, to

prevent that the mentioned deficiencies of the employed EOS adulterate the assessment.

To take into account that the precision of molecular simulation results varies in different regions, the deviation of each individual state point was put into relation with the average deviation of the data points in its neighborhood in the T – ρ plane. Each data point j has its own neighborhood with $i = 1, \dots, M$ neighbors, where the number of neighbors is $M = 15$ – 20 . The nearest neighbors were determined using the radial distance $\tilde{\delta}R$ between data points in the T – ρ plane, i.e., $\tilde{\delta}R = (\tilde{\delta}T^2 + \tilde{\delta}\rho^2)^{0.5}$. The choice of the radius and, hence, of M depends on the location of the data point in the T – ρ plane, i.e., in a fluid region where data points are sparse or if a data point is close to a phase boundary, the radius was chosen large enough such that at least 15 neighbors were allocated to the neighborhood of each data point. The neighbors of each data point j were taken from the entity of all data points N of a given property, i.e., the neighborhood M contains in general data points from different publications.

To decide whether a data point j is an outlier, a measure P_j is introduced. Following the ideas of refs 38 39 206, and 209, P_j was defined as

$$P_j = \frac{|\delta x_j - \text{median}_{i=1}^M(\delta x_{ji})|}{\text{MAD}_j}$$

with $i = 1 \dots M$ and $j = 1 \dots N$ (3)

In the measure P_j , the deviation of a tested data point j and an EOS are compared to the deviations of the data points from its neighborhood and that EOS. Details on the selection of the EOS are given below. The numerator on the right-hand side of eq 3 is the absolute deviation of the data point j and the median²¹⁰ of the data points in its neighborhood. The denominator is a measure for the mean absolute deviation (MAD) of the data points in the neighborhood of the given data point²¹¹ j

$$\text{MAD}_j = k \cdot \text{median}_{i=1}^M(|\delta x_{ji} - \text{median}_{i=1}^M(\delta x_{ji})|) \quad (4)$$

where $k = 1.4826$. The selection of that number is related to the assumption that the deviation data δx_{ji} are normally distributed in the set of data points.²¹² The MAD_j quantifies how strong the data points of the neighborhood M scatter on average from their median, i.e., how precise the data in the neighborhood are on average. The MAD is robust, as the median is less affected by outliers than the mean. Hence, this test takes the accuracy of the EOS and the accuracy of the computer experiment data of a certain thermophysical property in a certain fluid region into account. For brevity, it is called EOS test in the following.

The EOS test requires a sufficiently dense neighborhood of data points, to ensure that the accuracy of the reference, i.e., the EOS, remains fairly constant within that neighborhood. Since the data points for $T > 6$ are sparsely distributed in the T - ρ plane, the EOS test was only applied to data at $T \leq 6$.

The measure P_j was computed for the four EOS that were selected as described above and then compared with a parameter $P_{\max}$

$$P_j \geq P_{\max} \quad (5)$$

If this decision criterion was fulfilled for at least two of the four EOS, the data point j was identified as an outlier. The parameter $P_{\max}$ introduces an unavoidable subjective attribute into the method^{206,209} and regulates the severity of the EOS test. A fixed number for $P_{\max}$ was used here for all thermophysical properties, EOS, and fluid regions under consideration, which established a consistent framework. The parameter was set to $P_{\max} = 4$. This results in a confirmation rate of approximately 90% of all homogeneous data. Vice versa, 10% of all homogeneous data points were identified as outliers. This rather conservative choice^{206,209} for $P_{\max}$ was used to prevent a false assessment of data points to be erroneous. A more stringent, but smaller database can be generated easily by increasing $P_{\max}$ and vice versa, a larger database, which might however include less precise data, can be generated by decreasing $P_{\max}$. The EOS test enables a robust assessment of the quality of individual data points based on an assessment of the precision of the data points in its neighborhood.

Figure 2 summarizes the results of the EOS test for the homogeneous state points for $P_{\max} = 4$. About 85% of the pvT data and the data for the internal energy u passed the EOS test. These two thermophysical properties are the most frequently reported: the database contains about five times more pvT and internal energy data than any other thermophysical property. The relatively low confirmation rate of the pvT and internal

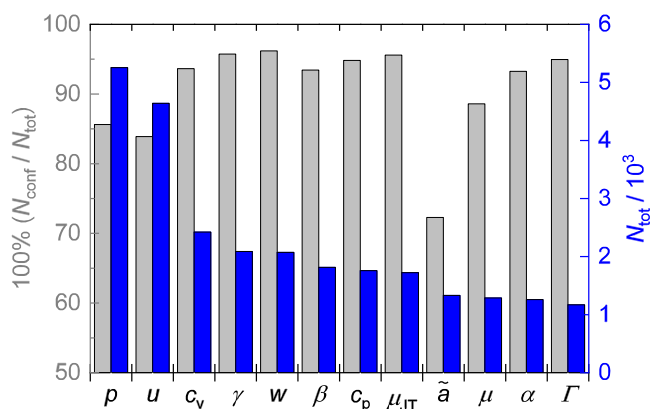


Figure 2. Statistics of the EOS test for homogeneous state points for different thermophysical properties: pressure p , internal energy u , isochoric heat capacity c_v , thermal pressure coefficient γ , speed of sound w , isothermal compressibility β , isobaric heat capacity c_p , Joule–Thomson coefficient μ_{JT} , Helmholtz energy divided by temperature $\tilde{a} = a/T$, chemical potential μ , thermal expansion coefficient α , and Grüneisen parameter Γ . The left ordinate indicates the percentage of confirmed data, and the right ordinate indicates the total number of data points available for each thermophysical property.

energy data is a result of significant differences in the precision among different publications that the EOS test used for the detection of the outliers.

Confirmation rates of 94–96% were obtained for data for the isochoric heat capacity c_v , isobaric heat capacity c_p , speed of sound w , thermal pressure coefficient γ , Joule–Thomson coefficient μ_{JT} , and Grüneisen parameter Γ —which are higher-order temperature and density derivatives of the Helmholtz energy. The confirmation rate for the thermal expansion coefficient α and isothermal compressibility β was 92–93%. A confirmation rate of 89% was found for the chemical potential μ , which is particularly challenging to determine by molecular simulation. The lowest confirmation rate of 72% was found for the Helmholtz energy $\tilde{a}$, which is mainly due to particularly low confirmation rates for the results from some publications. Details are given in the [Supporting Information](#).

Also, the numerical values of all data points and the corresponding result of the EOS test, i.e., whether a data point is confirmed or identified as an outlier with $P_{\max} = 4$, are provided in the [Supporting Information](#). Moreover, information on the dependence of the total percentage of confirmed data on the choice of the number for $P_{\max}$ is presented there.

Assessment of Vapor–Liquid Equilibrium Data. For the vapor–liquid equilibrium, both bulk and interfacial properties were considered: vapor pressure p^s , saturated liquid and vapor densities ρ' and ρ'' , respectively, and enthalpy of vaporization Δh_v were considered for the bulk properties and the surface tension γ as interfacial property (cf. [Table 1](#)).

Bulk VLE Data. In contrast to the test on the homogeneous state points, which was assessed for each data point individually, entire VLE bulk property data sets from a given publication were assessed and either confirmed or discarded. Furthermore, the EOS test for homogeneous data is designed to identify gross outliers, whereas the assessment of the VLE data additionally aims at determining the most precise data and to give an estimation of that precision. The estimated precision indicates the magnitude that a given data set scatters around the presumed true value. The true value, i.e., exact and correct, of a given property is approximated here by the mean of the most precise and accurate data sets.

Table 3. Parameters of Correlations 6–9 for the Vapor Pressure p^s , Saturated Liquid Density ρ' , Saturated Vapor Density ρ'' , and Enthalpy of Vaporization Δh_v

	p^s (eq 6)		ρ' (eq 7)		ρ'' (eq 8)		Δh_v (eq 9)	
	n_i		n_i	t_i	n_i	t_i	n_i	t_i
1	1.156551		1.341700	0.327140	−8.135822	1.651685	6.456728	0.411342
2	−4.431519		2.075332	0.958759	−102.919110	43.469214	2.700099	0.460416
3	−0.423028		−2.123475	1.645654	−3.037979	0.462877	−3.073573	2.350953
4	2.638743		0.328998	17.000001	−44.381841	11.500462	3.149052	5.017010
5			1.386131	2.400858	−34.55892948	5.394370		

VLE bulk data of the Lennard-Jones fluid were assessed by several independent consistency tests. Two of them were taken from the literature: the compressibility factor test proposed by Nezbeda^{213,214} and the Clausius–Clapeyron test.⁷¹ Furthermore, as a third test, outliers were determined from a direct comparison of the data points as described below in more detail. For brevity, this test is referred to as deviation test in the following. Data sets were discarded, if the majority of data points within a data set violate one or more consistency tests. For the following discussion, the vicinity to the critical point of the VLE region is defined to be above 95% of the critical temperature. Data points in that region were not included in the consistency tests.

To facilitate the assessment, empirical correlation functions for the vapor pressure p^s , saturated liquid and vapor densities ρ' and ρ'' , respectively, and enthalpy of vaporization Δh_v were used

$$\ln p^s = n_1 T + \frac{n_2}{T} + \frac{n_3}{T^{n_4}} \quad (6)$$

$$\frac{\rho'}{\rho_c} = 1 + \sum_{i=1}^5 n_i \left(1 - \frac{T}{T_c} \right)^{t_i} \quad (7)$$

$$\ln \left(\frac{\rho''}{\rho_c} \right) = \sum_{i=1}^5 n_i \left(1 - \frac{T}{T_c} \right)^{t_i} \quad (8)$$

$$\Delta h_v = \sum_{i=1}^4 n_i (T_c - T)^{t_i} \quad (9)$$

In an initial step, the VLE data set from this work was used for the parametrization of eqs 6–9, as it is the most extensive data set. The absolute average deviation from the correlations 6–9 and the respective VLE data from this work are: 0.3% for the vapor pressure, 0.06% for the saturated liquid density, 0.5% for the saturated vapor density, and 0.3% for the enthalpy of vaporization. The numerical values of the parameters n_i and t_i are listed in Table 3. The critical values used in eqs 6–9 are $T_c = 1.321$ and $\rho_c = 0.316$. They are called base correlations in the following. It will be shown below that these correlations match the best available guess for the considered properties within their estimated precision.

In the test proposed by Nezbeda,^{213,214} the compressibility factor of the saturated vapor phase $Z'' = p^s/\rho''T$ is considered. Starting at low temperatures close to the triple point, Z'' must be close to unity. With increasing temperature, Z'' must decrease monotonically until the compressibility factor at the critical point is reached.

This simple criterion can be applied for sorting out outliers: $Z'' > 1$ is not acceptable and the slope dZ''/dT must be negative. More generally, the slope dZ''/dT is sensitive and

useful for discriminating data points that deviate from the general trend. Furthermore, the data points were compared to the results for Z'' , which was obtained from the base correlations. Data points for which the deviation in Z'' from that correlation is above 1.25% were considered as outliers. The number 1.25% was chosen on the basis of an examination of the scattering of all available data. Data sets of which more than 50% of the data points violate at least one of these criteria were discarded. The other data sets were considered as confirmed by the compressibility factor test.

Figure 3 shows the data sets that were confirmed by the compressibility factor test. These data are in good mutual

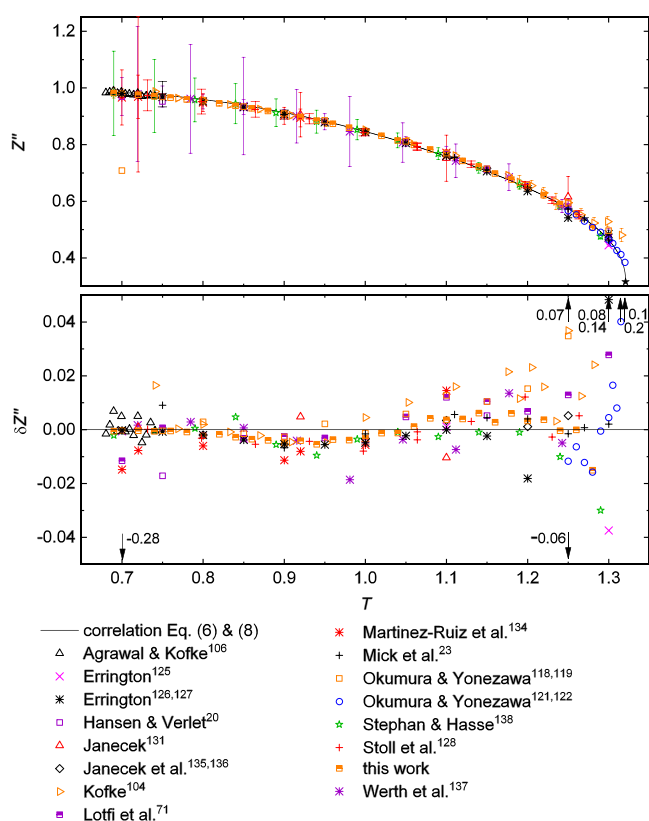


Figure 3. Compressibility factor test for the vapor–liquid equilibrium data of the Lennard-Jones fluid: saturated vapor-phase compressibility factor Z'' as a function of the temperature T (top) and the relative deviation of Z'' from the base correlations 6 and 8 (bottom). The black filled star indicates the compressibility factor at the critical point according to the base correlation. Error bars are omitted in the bottom plot to avoid visual clutter.

agreement and in agreement with the base correlation computed from eqs 6 and 8. Excellent mutual agreement is found for the data sets from Agrawal and Kofke,¹⁰⁶ Errington,¹²⁵ Errington,^{126,127} Janeček et al.,^{135,136} Lotfi et al.,⁷¹ Mick et al.,²³

Okumura and Yonezawa,^{121,122} and the present work. Taking these data as reference, and their scattering as a measure for the precision, with which Z'' is known, precision is estimated to be approximately $\pm 0.5\%$, excluding the vicinity of the critical point. Furthermore, the data sets from Hansen and Verlet,²⁰ Janeček,¹³¹ Kofke,¹⁰⁴ Martínez-Ruiz et al.,¹³⁴ Okumura and Yonezawa,^{118,119} Stephan and Hasse,¹³⁸ Stoll et al.,¹²⁸ and Werth et al.¹³⁷ also agree well with the data mentioned above, but scatter more. These data were confirmed by the compressibility factor test. The larger scatter of the data of refs 134, 137, and 138 is probably due to the fact that these data were sampled with the direct simulation method. The data sets from refs 118, 119, and 125–127 each contain a single data point that is a clear outlier. These data points are explicitly pointed out in the [Supporting Information](#). However, apart from the outlier, these data sets are considered as confirmed by the compressibility factor test. Data sets from refs 58, 60, 100, 101, 103, 107, 108, 110, 113–115, 117, 124, 129, 130, 215, and 216 were discarded by the compressibility factor test. Details are given in the [Supporting Information](#).

The second consistency test that was employed for the VLE bulk data is based on the Clausius–Clapeyron equation

$$\frac{d \ln p^s}{d(1/T)} = - \frac{T \Delta h_v}{p^s(1/\rho'' - 1/\rho')} \quad (10)$$

The left-hand side (LHS) and right-hand side (RHS) of eq 10 are considered individually for each data set and compared. The equality of the LHS and RHS within the corresponding statistical uncertainties indicates the thermodynamic consistency by the Clausius–Clapeyron equation. The RHS of eq 10 was therefore computed directly from the numerical values of the computer experiments. The statistical uncertainties of the RHS values were determined using the error propagation law. The LHS of eq 10 was calculated with an analytical function for the vapor pressure curve. The Clausius–Clapeyron test was applied here in two ways (test A and test B). In test A, the LHS and RHS of a given data set were directly compared. This can only be done in a meaningful way for those data sets that enables the parametrization of a correlation of $p^s(T)$ for the LHS in a sufficiently accurate manner. In test B, the RHS correlations of remaining data sets are compared with the LHS correlations established in test A. Data points violate the Clausius–Clapeyron equation in test A, if the computed RHS of a given data point does not agree within the statistical uncertainty with the corresponding LHS value. In test B, data points were considered as outliers, if they deviate 2% or more from the two correlations established in test A. As for the compressibility factor test, data sets for which more than half of the data points violate the Clausius–Clapeyron test were discarded. The remaining data sets were considered as confirmed by the Clausius–Clapeyron test.

The data sets from Lotfi et al.⁷¹ and this work were employed in test A. Correlations for $p^s(T)$ obtained from the other data sets were not found to be sufficiently accurate for carrying out a meaningful test of type A. For the data from this work, the base correlation for $p^s(T)$ from eq 6 was used for comparison. The statistical uncertainty of the LHS was estimated from the error propagation law. Lotfi et al.⁷¹ published a correlation of their own p^s results that was used here for computing the LHS of eq 10 for the comparison of the corresponding RHS values.

Figure 4 shows the VLE data sets that were confirmed by both the Clausius–Clapeyron test and the compressibility

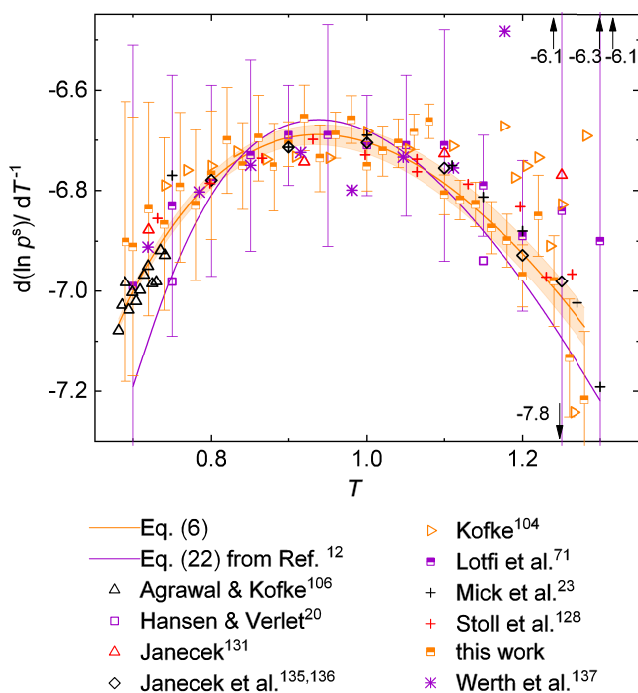


Figure 4. Clausius–Clapeyron test according eq 10. The symbols indicate the RHS, and the lines the LHS of eq 10. The orange shaded area indicates the statistical uncertainty of $\pm 2\%$ computed for the LHS of eq 6. The purple line is eq 22 from ref 71.

factor test. In p^s results plot, results for the RHS of eq 10 that were obtained from the simulation data and are shown as symbols are compared to the results for the LHS that were obtained from correlations of the vapor pressure curve and are shown as lines. Two such correlations are depicted: one obtained from the present work and the other obtained by Lotfi et al.⁷¹ Lotfi et al.⁷¹ compared their results for the Lennard-Jones fluid with results for Ne, Ar, and CH₄ and found very good qualitative agreement, i.e., a parabolic shape of $d(\ln p^s)/d(T^{-1})$.

Figure 4 shows that the two correlations of the LHS of eq 10 match well. The differences are far below the uncertainties of the data points for the RHS of eq 10. The data from the present work and from Lotfi et al.⁷¹ agree well with the correlations. There is one outlier in the data set from the present work. (It was not removed from the data set to ensure a fair comparison with the literature data, for which almost every data set was found to have at least one outlier.) The LHS and RHS of eq 10 agree within the statistical uncertainties for all but this state point, which is likely the result of an overly optimistic error estimation. However, the data of Lotfi et al.⁷¹ and this work agree very well throughout, but the data of Lotfi et al.⁷¹ show a more pronounced scatter and larger error bars than the data from this work.

The RHS data of Janeček et al.,^{135,136} Stoll et al.,¹²⁸ Mick et al.,²³ and Agrawal and Kofke¹⁰⁶ are also in excellent agreement with the base correlation of the LHS of eq 10. The agreement is also very good for the data sets of Hansen and Verlet,²⁰ Janeček,¹³¹ Kofke,¹⁰⁴ and Werth et al.,¹³⁷ but their RHS data exhibit considerable scatter, as does the data set of Lotfi et al.⁷¹

The data sets from refs 58, 60, 100, 101, 103, 108, 110, 124, and 130 were discarded by the Clausius–Clapeyron test. Details are given in the [Supporting Information](#). The results

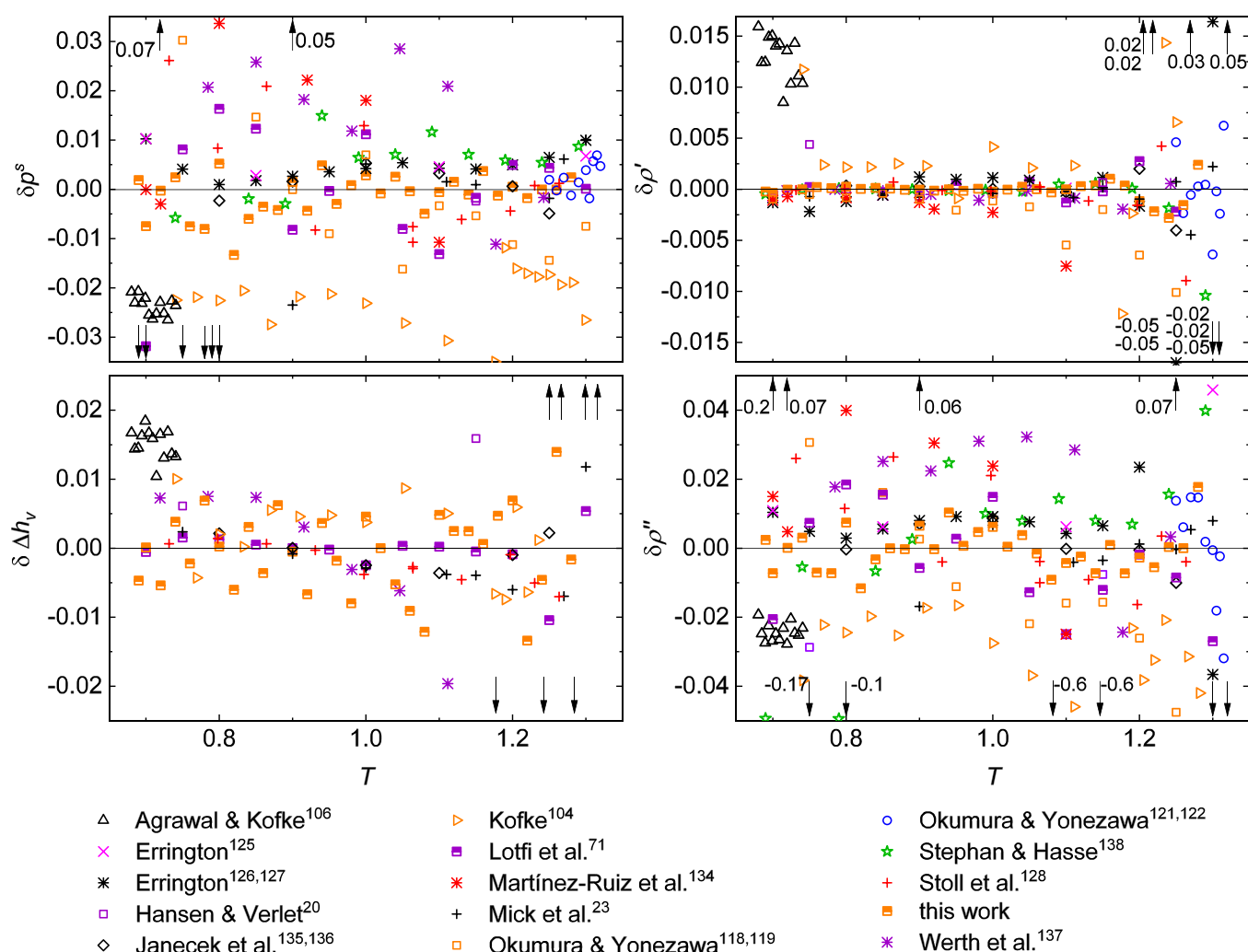


Figure 5. Relative deviations of the vapor–liquid equilibrium data for the vapor pressure p^s , saturated liquid density ρ' , saturated vapor density ρ'' , and enthalpy of vaporization Δh_v from correlations 6–9 as a function of the temperature T . Error bars were omitted to avoid visual clutter. For clarity, numerical values for out-of-range data points with negative deviation for the vapor pressure, the enthalpy of vaporization, and the saturated vapor density ρ'' are omitted in the respective plot; they lie in the range of $\delta p^s = -0.05$ to -0.16 , $\delta \Delta h_v = -0.15$ to 0.45 , and $\delta \rho'' = -0.08$ to -0.2 .

from the Clausius–Clapeyron test reinforce the findings from the compressibility factor test.

As a third assessment, deviation plots of each VLE bulk property (p^s , ρ' , ρ'' , and Δh_v) from the base correlations were used as follows: First, the data sets with the best mutual agreement were identified that were also confirmed by the previous two tests. From these data sets, a confidence interval $\pm X\%$ was estimated for each VLE property ($X = p^s$, ρ' , ρ'' , and Δh_v) individually that is a measure for the precision of the available information. A data point was considered as an outlier, if its relative deviation to the corresponding base correlation exceeds 2.5 times the number of X of one of the four VLE properties. Again, data sets for which more than half of the data points of a given VLE property violate the criterion were discarded. The remaining data sets were considered as confirmed.

Figure 5 shows the relative deviation of the VLE data (vapor pressure p^s , enthalpy of vaporization Δh_v , and the saturated densities ρ' and ρ'') from the base correlations. Excellent mutual agreement (and also agreement with the base correlations) was found for the data sets from Errington,¹²⁵ Errington,^{126,127} Janeček et al.,^{135,136} Lotfi et al.,⁷¹ Mick et al.,²³ Okumura and Yonezawa,^{121,122} and the data set from the

present work. These seven data sets were also confirmed by the other two tests and comprise a total of 86 state points. Interestingly, the precision, i.e., the scatter, of each of the seven best data sets (refs 23, 71, 121, 122, 125–127, 135, 136 and this work) is very similar. Only the saturated vapor density and vapor pressure data of Errington^{126,127} were found to be significantly more precise than the other data sets. However, the scatter of the saturated liquid density data of Errington^{126,127} is significantly larger than the scatter of the other selected best data sets (refs 23, 71, 121, 122, 125–127, 135, and 136 and this work).

The mutual agreement of these seven selected best bulk VLE data sets is approximately $\pm 1\%$ for the vapor pressure, $\pm 0.2\%$ for the saturated liquid density, $\pm 1\%$ for the saturated vapor density, and $\pm 0.75\%$ for the enthalpy of vaporization—excluding the region close to the critical point. Five of these seven best data sets contain a single or two clear outliers, which are listed in the Supporting Information.

The mutual agreement of the seven selected best data sets (refs 23, 71, 121, 122, 125–127, 135, and 136 and the data set from the present work) significantly decreases in the region close to the critical point, i.e., $T \gtrsim 1.25$. The mutual agreement

in the vicinity of the critical point is better for the vapor pressure and the saturated liquid density than for the enthalpy of vaporization and the saturated vapor density.

The following data sets agree well but scatter strongly (however, within the confidence interval and are thereby considered as confirmed by the deviation test): Betancourt-Cárdenas et al.,¹³⁰ Chen et al.,²⁴ Hansen and Verlet,²⁰ Martínez-Ruiz et al.,¹³⁴ Okumura and Yonezawa,^{118,119} Stoll et al.,¹²⁸ Stephan and Hasse,¹³⁸ and Werth et al.¹³⁷ The data sets from refs 58, 60, 98–115, 117, 123, 124, 129, 131–133, and 197 were discarded because more than 50% of the respective data points exceed the confidence interval of one of the bulk VLE properties.

The seven data sets from refs 23, 71, 121, 122, 125–127, 135, and 136 and from the present work were confirmed by all applicable tests and were found to be in excellent mutual agreement and are therefore recommended as a reference. The data sets from refs 20, 118, 118, 128, 134, 137, and 138 were also confirmed by all applicable tests but exhibit a significantly larger scattering compared to these seven most precise data sets. The data sets from refs 24, 98, 102, 105–107, 111, 112, 116, 120, 123, and 132 were discarded by a single consistency test, the data sets from refs 104, 113–115, 117, and 129 were discarded by two tests, and the data sets from refs 60, 100, 101, 103, 124, and 130 were discarded by three tests. Note that, for most data sets, only a selection of tests could be applied due to missing properties that were not reported in these publications. For all but one VLE data set, the results of the three applicable tests reinforce each other. The only exception is the data set of Agrawal and Kofke:¹⁰⁶ the data set was confirmed by the compressibility factor test and the Clausius–Clapeyron test but discarded by the deviation test.

For many of the investigated data sets, the data points do not agree with the seven selected best data sets within the reported statistical uncertainties. Therefore, systematic simulation errors²⁹ are held responsible for these deviations.

Fitting the combined data of the seven best data sets (refs 23, 71, 121, 122, 125–127, 135, and 136 and the data set from the present work) with the correlations presented in eqs 6–9 yields a fit that is essentially the same as the one presented in Table 3. For all studied properties, the differences between both fits are well below the differences among the data from the seven data sets. Hence, we simply recommend using the correlations from Table 3.

Interfacial VLE Data. As for the bulk VLE properties, the computer experiment data for the surface tension were assessed with respect to their mutual agreement. To facilitate this assessment, a base correlation function was also used for the temperature dependence of the surface tension

$$\gamma = A \left(1 - \frac{T}{T_c} \right)^b \quad (11)$$

where A and b are parameters. $T_c = 1.321$ was specified for the critical temperature in eq 11. In an initial step, the surface tension data from refs 138 and 142 were used for the parametrization of eq 11. The obtained values for the parameters are: $A = 3.03327$ and $b = 1.27748$. The absolute average deviation between correlation 11 and the data used for the parametrization is 1.2%. It will be shown below that eq 11 matches the best available guess for the surface tension within its estimated precision.

The surface tension data were also assessed with the deviation test: First, the data sets with the best mutual agreement

were identified. From these data sets, a confidence interval $\pm X\%$ was estimated as a measure for the precision that the surface tension is known. For the assessment, a data point was considered as an outlier, if its relative deviation to the base correlation 11 exceeds 2.5 times the number of $\pm X\%$. Again, data sets for which more than half of the data points violate the criterion were discarded. The remaining data sets were considered as confirmed.

Figure 6 shows the surface tension data as a function of the temperature. Overall, the agreement is significantly poorer than

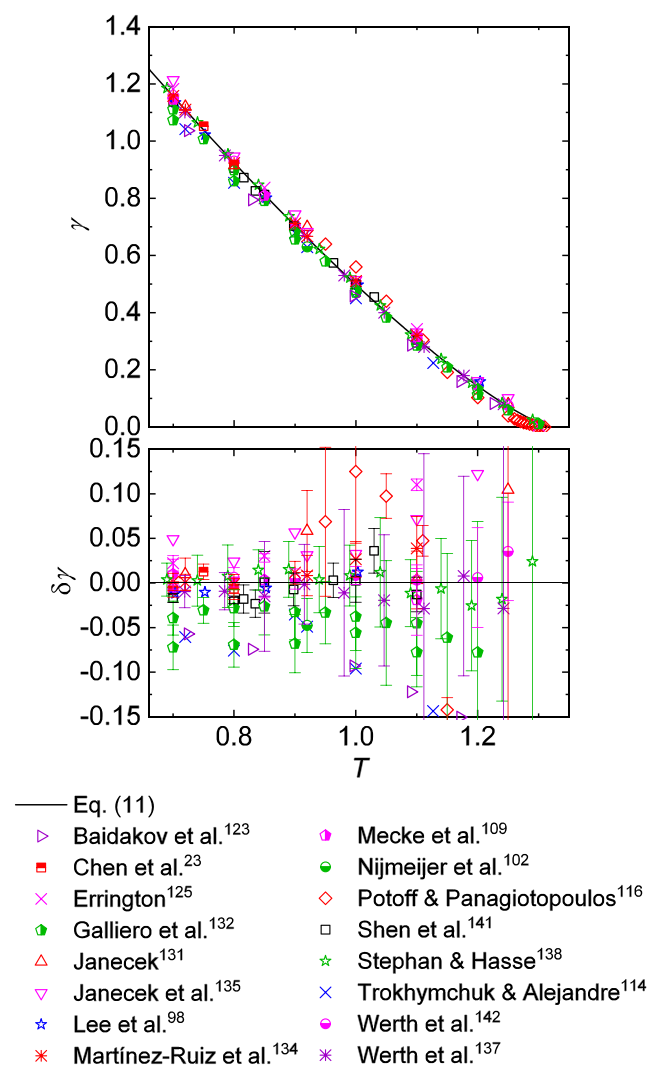


Figure 6. Surface tension of the Lennard–Jones fluid as a function of the temperature (top) and the corresponding relative deviation plot (bottom). The black line (top) and baseline (bottom) indicate eq 11. The symbols indicate computer experiment data. For clarity, numerical values for out-of-range data points in the vicinity of the critical temperature are omitted in the deviation plot.

for the bulk VLE properties. The best mutual agreement is found for the data sets of Chen et al.,²⁴ Janeček et al.,¹³⁵ Janeček,¹³¹ Werth et al.,¹⁴² Werth et al.,¹³⁷ Lee et al.,⁹⁸ Martínez-Ruiz et al.,¹³⁴ Mecke et al.,¹⁴⁰ Nijmeijer et al.,¹⁰² Stephan and Hasse,¹³⁸ and Shen et al.¹⁴¹ These data sets agree approximately within $\pm 4\%$ —excluding the region close to the critical point.

The following data sets agree well, i.e., are considered as confirmed, but exhibit a more pronounced scatter or systematic

deviations from the aforementioned data sets: Trokhymchuk and Alejandro,¹¹⁴ Potoff and Panagiotopoulos,¹¹⁶ Miyazaki et al.,¹³⁹ Galliero et al.,¹³² Errington,¹²⁵ Baidakov et al.,¹²⁹ and Baidakov et al.¹²³ The data sets of refs 99, 105, 111, 112, 115, and 117 were discarded since more than 50% of the respective data points exceed the confidence interval. Details are given in the [Supporting Information](#).

CONCLUSIONS

This work reviews and assesses thermophysical property data for the Lennard-Jones fluid. Literature data (approximately 35 000 data points) were digitalized, evaluated, and provided in the [Supporting Information](#) as a consistent database. For the homogeneous state points, approximately 10% of the data were identified to be gross outliers by an EOS test. The lack of information on the statistical uncertainties in some studies is unfortunate.

Three independent consistency tests were employed for the evaluation of the VLE data: The compressibility factor test considers only the vapor pressure and the saturated vapor density, whereas the Clausius–Clapeyron test also considers the saturated liquid density and the enthalpy of vaporization. The third test evaluates the mutual agreement of data sets for each VLE property individually. The VLE data sets of Errington,¹²⁵ Errington,^{126,127} Janeček et al.,^{135,136} Lotfi et al.,⁷¹ Mick et al.,²³ Okumura and Yonezawa,^{121,122} and this work were confirmed by all tests and found to be the most precise. The precision with which the VLE of the Lennard–Jones fluid is known is thereby estimated to be $\pm 1\%$ for the vapor pressure, $\pm 0.2\%$ for the saturated liquid density, $\pm 1\%$ for the saturated vapor density, and $\pm 0.75\%$ for the enthalpy of vaporization—excluding the region close to the critical point. Since almost every VLE data set—even the most precise—contains individual outliers, we recommend to use the combination of the above listed data sets as reference. Furthermore, the data sets of Hansen and Verlet,²⁰ Martínez-Ruiz et al.,¹³⁴ Okumura and Yonezawa,^{118,119} Stephan and Hasse,¹³⁸ Stoll et al.,¹²⁸ and Werth et al.¹³⁷ were found to be accurate but not as precise. The precision of the surface tension is estimated to be $\pm 4\%$, i.e., significantly larger than the VLE bulk properties.

The EOS test presented here can be adopted easily to other applications in thermodynamics. A test based on the ideas described here could, for example, be used for testing VLE data of a given mixture based on models of the excess Gibbs energy G^E .

In a follow-up work, we will use the consolidated database that is presented here to investigate the performance of equations of state for the Lennard–Jones fluid. Future work should focus on the region in the direct vicinity of the critical point, where the available VLE data and accordingly the location of the critical point agree less well.

Among the molecular model fluids, the Lennard–Jones fluid plays a similar role to water among real fluids: it is the fluid that has been studied most extensively and most thoroughly. The present work summarizes the present state of these studies. The accuracy with which the properties of the Lennard–Jones fluid are known is assessed here. It can be assumed that it is a lower limit to the accuracy with which properties of fluids can presently be determined with molecular simulations. That accuracy often exceeds the statistical uncertainties that are reported for individual data points.

Appendix: Molecular Simulations of This Work

Homogeneous State Point Simulations. Simulations in the homogeneous state were carried out in the *NVT* ensemble

using Monte Carlo (MC) sampling with an acceptance ratio of 0.5. The simulations contained 1372 particles. The initial conditions were generated as a face-centered cubic (fcc) lattice and subsequently equilibrated for 2.5×10^5 cycles and then sampled for 3×10^6 cycles, where one cycle corresponds to 1372 attempts for a translational move. The cutoff radius was half the edge length of a cubic simulation volume. Statistical uncertainties were estimated by the block averaging method of Flyvbjerg and Petersen.³⁴ Numerical values of the simulation results are reported in the [Supporting Information](#). Simulations were carried out with the simulation program *ms2*.¹⁷⁷

VLE Simulations. The vapor–liquid equilibrium of the Lennard–Jones fluid was obtained with the Grand Equilibrium method of Vrabec and Hasse.¹⁷⁹ Two simulations were subsequently carried out to obtain the VLE properties at a given temperature: First, the chemical potential and the volume were sampled in an *NpT* MD liquid-phase simulation. Then, the vapor phase was simulated in a pseudo- μVT ensemble using an MC algorithm. The pressure of the liquid state was chosen to be approximately 7% above the vapor pressure, as estimated by the Lennard–Jones EOS from ref 207. 2048 particles were initialized on an fcc lattice for the *NpT* simulations and then equilibrated for 3×10^5 time steps. The time step was set to $\Delta\tau = 0.001$. Velocity scaling was used to specify the temperature. The chemical potential was sampled with Widom’s test particle method,¹⁷⁶ inserting 2500 test particles every time step. The chemical potential and its pressure derivative, i.e., the volume, were then used during the second simulation to obtain the saturated vapor-phase properties. The pseudo- μVT simulation was carried out such that the specified chemical potential was not constant but a function of the actual pressure of the vapor phase. The vapor-phase simulations were started with 1000 particles using MC sampling with an acceptance ratio of 0.5. The system was equilibrated for 10^5 time cycles, and production was carried out for 5×10^5 time cycles. The initial vapor density was also estimated with the Lennard–Jones EOS from ref 207. As for the homogeneous state point simulations, the cutoff radius was half the edge length of a cubic simulation volume and the statistical uncertainty was estimated by the block averaging method.³⁴ The vapor–liquid equilibrium was sampled at 31 temperatures between $T = 0.69$ and 1.28. Numerical values of the obtained vapor pressure, saturated densities, and enthalpy of vaporization are reported in the [Supporting Information](#).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jcim.9b00620.

Detailed remarks on individual data sets report on conversion pitfalls, likely misprints, etc.; results of all three VLE tests for the discarded VLE data sets and details of the EOS test ([PDF](#))

Database as an .xls spreadsheet of all discussed thermophysical property data in this work: each data point is listed with the reference of the original publication, temperature, density, the property itself, eventually the reported statistical uncertainty, and the result of the EOS test; all data are given in standard Lennard–Jones reduced units ([XLSX](#))

AUTHOR INFORMATION

Corresponding Author

*E-mail: simon.stephan@mv.uni-kl.de

ORCID

Simon Stephan: 0000-0002-4578-3569

Notes

The authors declare no competing financial interest.

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